AP PRECALCULUS

Unit 1 • Worksheet 1.11A • Answer key

Worksheet 1.11A • Rational Functions and Concept of Limit

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$\lim_{x \to 3} \dfrac{x^2 - 3x}{x - 3}$ is

(A) 0

(B) 1

(C) 9

(D) 3

Solution

Answer: (D). $\frac{x(x-3)}{x-3} = x \to 3$.

Question 2

NO CALCULATOR

$\lim_{x \to 2} \dfrac{x^2 - x - 2}{x^2 - 4}$ is

(A) $\frac{3}{4}$

(B) 0

(C) $\frac{1}{2}$

(D) It does not exist.

Solution

Answer: (A). $\frac{(x-2)(x+1)}{(x-2)(x+2)} \to \frac{3}{4}$.

Question 3

NO CALCULATOR

$\lim_{x \to -1^-} \dfrac{x+3}{x+1}$ is

(A) ∞

(B) $-\infty$

(C) 2

(D) 1

Solution

Answer: (B). Numerator near 2; for $x < -1$, $x + 1 < 0$.

Question 4

NO CALCULATOR

Which describes $\lim_{x\to 4} \dfrac{1}{x-4}$?

(A) ∞
(B) $-\infty$
(C) 0
(D) It does not exist (the one-sided limits are $-\infty$ and ∞).

Solution
Answer: (D). Opposite infinite one-sided limits.

Question 5 (Free response)

NO CALCULATOR

Let $r(x) = \dfrac{2x^2+2x-4}{x^2-1}$. Describe each feature with a limit statement: (a) the hole, (b) the vertical asymptote, (c) the end behavior.

Solution
$r = \frac{2(x+2)(x-1)}{(x-1)(x+1)}$. (a) $\lim_{x\to 1} r(x) = 3$ (hole at $(1,3)$). (b) $\lim_{x\to -1^-} r = -\infty$, $\lim_{x\to -1^+} r = +\infty$. (c) $\lim_{x\to \pm\infty} r = 2$.

Question 6 (Free response)

NO CALCULATOR

Find each limit, or explain why it does not exist. (a) $\lim_{x\to -2} \dfrac{x^2+5x+6}{x+2}$ (b) $\lim_{x\to 0^+} \dfrac{x-1}{x^2}$ (c) $\lim_{x\to\infty} \dfrac{3-x^2}{2x^2+x}$ (d) $\lim_{x\to 1} \dfrac{x}{x-1}$

Solution
(a) $x+3 \to 1$. (b) numerator near -1, $x^2 > 0$: $-\infty$. (c) $-\frac{1}{2}$. (d) Does not exist: $-\infty$ from the left, $+\infty$ from the right.